

Basis-Guided Counterfactual Generation for Explainable Multivariate Time Series Models

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Abstract. Explainability in time series is critical for establishing trust in deep learning models. While counterfactual explanations offer a path to understanding model decisions by suggesting minimal changes to input data, that is, "what would need to change for the model to predict a different outcome", existing approaches often treat time-series data as independent features and optimise the perturbations directly over it. In the case of multivariate sensor data, this often yields jittery, high-frequency changes that can satisfy the model, but the counterfactuals generated physically might not be plausible. To address this, we propose a novel method that parameterises counterfactual perturbations using smooth temporal basis functions (e.g., B-Splines, Fourier) rather than raw time steps. By optimising weights in a low-dimensional basis space, it inherently enforces temporal coherence, eliminating unrealistic noise without requiring complex physics-based constraints. We further introduce a diversity mechanism based on Determinantal Point Processes (DPPs) in the coefficient space to generate distinct, valid counterfactual trajectories. To evaluate the proposed method, we used various multivariate time series datasets and generated counterfactuals. We compared our proposed approach with the state-of-the-art methods; our approach consistently produces valid counterfactuals while achieving a better trade-off between proximity, sparsity, and temporal smoothness than existing baselines.

Keywords: Explainable AI · Basis Functions · Counterfactual Explanation · Multivariate Time Series.

1 Introduction

With recent advancement in Machine Learning, we see that deep learning models perform good and had become a standard for complex prediction tasks. While these models being adopted and deployed in the high-stake domains these models lacks limited interpretability and explainability[33]. This lack of transparency is risky in high-stakes settings like healthcare, industrial monitoring, finance, and autonomous decision support, where interpretability is closely tied to trust, accountability, and safe adoption. Recent reviews of Explainable Artificial Intelligence (XAI) continue to identify explanation quality, robustness, and evaluation as open challenges for real-world deployment[20,22].

A large part of the XAI literature has focused on feature-attribution and local surrogate methods[10], such as LIME[25] and SHAP[21], which explain predictions by estimating the contribution of input variables. These methods are useful for highlighting which parts of an input were influential (feature importance) and for explaining *why* the model predicted a given outcome and which feature contributed the most. But, these models do not show *How* the outcome could be changed. This can be tackled using counterfactual explanations, which provide alternative inputs that are close to the original instance yet lead to the desired prediction. Since the influential work of Wachter et al. [29], counterfactual explanations have become a prominent paradigm because they are naturally aligned with recourse, intervention, and actionability.

Despite rapid progress, most counterfactual methods were originally developed for tabular data [12]. The time-series setting is considerably more difficult because sequential data exhibit temporal dependence, local continuity [34], and, in multivariate settings, strong cross-channel interactions. These methods, naively perturb individual time points may flip the model output while producing unrealistic, noisy, or physically implausible trajectories. This limitation has been repeatedly noted in the emerging literature on XAI for time series, which remains far less mature than its tabular counterpart. Early and recent methods such as Glacier [31] have demonstrated the promise of time-series-specific counterfactual generation, but they also highlight the difficulty of balancing validity, sparsity, and temporal coherence.

In this paper we propose our novel approach to generate Counterfactuals for multivariate time series through a temporal basis formulation. Instead of optimising unrestricted perturbations over raw time steps, the proposed method parameterises counterfactual changes in a low-dimensional basis space using smooth temporal basis functions such as B-splines or Fourier bases. This embeds temporal coherence directly into the search space and helps generate explanations that are not only valid and proximal, but also smoother, more plausible, and easier to interpret as realistic alternative trajectories. In addition, diversity is encouraged in the coefficient space to produce multiple non-redundant counterfactual explanations for the same query instance.

2 Related Work

Most explainability methods are post-hoc, where we do not open the black box. Local Interpretable Model-agnostic Explanations (LIME) [25] explains an individual prediction by sampling perturbations around the instance and fitting an interpretable model locally, giving a faithful-but-local approximation of the decision surface. SHapley Additive exPlanations (SHAP) [21] provides an explanation from a game theoretic approach by assigning each feature an importance value for a particular prediction by the black-box model. When explaining models trained on time series data using a tabular approach, they can generate out-of-distribution subsequences and ignore temporal dependence [13]. This has motivated the development of various explainability methods explicitly for

time series scenarios, such as TimeSHAP [5], which uses sequence perturbations, and KernelSHAP [21], which extends KernelSHAP to sequential data to explain Recurrent Neural Networks (RNNs). TIMING [15] uses conventional Integrated Gradients (IG) [28] for feature attributions.

Counterfactual explanations are one of the most important topics in the field of explainable AI because they move beyond descriptive importance scores and instead provide actionable explanations: they indicate how an input would need to change in order to obtain a different prediction. This has deep roots from Structural Causal Models (SCM) by Judea Pearl, which formalises counterfactual queries via interventions and structural equations, setting a principled foundation for "what-if" reasoning [14]. In [29], Wachter et al. popularised counterfactual explanations (CFEs) for black-box automated decisions by framing them as minimal changes that would flip an outcome without exposing model internals. In [23] DiCE was introduced where CFEs are generated where diverse as it used determinantal point processes (DPP) objective.

Although counterfactual explanation is now well developed for tabular data, the time-series setting remains much less mature. This is mainly because sequential data introduce temporal dependence, high dimensionality, and strong structural constraints: changing isolated values may flip a prediction while producing unrealistic trajectories. In [9] the authors proposed "Native Guide" that generates CFEs by adapting discriminative regions toward a target-class guide. Glacier [32], a model-agnostic method designed specifically for generating locally-constrained counterfactual explanations for univariate time series classification, leveraging gradient search in either the original or a latent space learned via an auto-encoder. In [4] DiscoX, a Discord-based Counterfactual Explanation algorithm, as a post-hoc, model-agnostic method for time series classification models. In [24] Sub-SpaCE uses a genetic algorithm to enforce sparsity and contiguity via subsequence edits. forecasting CFEs were explicitly formulated in ForecastCF [30], and the input data is gradually adjusted a little at a time using gradient-based optimisation. In [3], CoMTE was introduced, where the CFEs were generated for multivariate time series data by finding similar examples from the training set that already give the desired output, and it then swaps a small number of variables from those examples into the current samples until the prediction changes.

To summarize the existing methodologies, we can see that most of the CFE generation methods performs well and uses various techniques like, nearest-neighbour, optimization in the latent space. Generating multivariate Counterfactuals is challenging, as preserving inter-channel correlation and temporal smoothness while ensuring validity and sparsity is difficult. Therefore, we generate our CFE on a smooth basis, which yields a non-redundant solution.

3 Methodology

Let the input be a time-series instance $\mathbf{X} \in \mathbb{R}^{T \times D}$, where $T \in \mathbb{N}$ is the sequence length and D is the number of sensor features. The input instance $X \in \mathbb{R}^{T \times D}$ be

the query instance(the original time series data) for which we seek an explanation for the data be,

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T]^\top, \quad \text{where } \mathbf{x}_t \in \mathbb{R}^D, t \in \{1, 2, \dots, T\} \quad (1)$$

We train a predictive model like Transformer or LSTM, let it be

$$f : \mathbb{R}^{T \times D} \rightarrow \mathbb{R} \quad (2)$$

We train the model as defined in Equation 2 to predict the $\hat{y}$ for example RUL(remaining Useful Life) of a machine/engine

$$\hat{y} = f(\mathbf{X}) \in \mathbb{R}^+ \quad (3)$$

3.1 Counterfactual Explanations

Counterfactual explanations answer the question: "What is the minimal change to the input features that would result in a desired different outcome?" In our context, this translates to: "What are the smallest changes in the sensor reading history that would result in the RUL prediction falling within a desired range?". We seek a counterfactual instance $\mathbf{X}_{\text{cf}}$ such that $f(\mathbf{X}_{\text{cf}}) \approx y_{\text{target}}$, where y_{target} is the Target value we want to achieve by the model f , for that we would do a minimal change to the original $\mathbf{X}$

$$\mathbf{X}_{\text{cf}} = \mathbf{X} + \mathbf{\Delta} \quad \text{where } \mathbf{\Delta} \in \mathbb{R}^{T \times D} \quad (4)$$

From Equation 4 we define $\mathbf{\Delta}$ as a free parameter matrix, the $\mathbf{\Delta}$ is constrained to lie in the span of a smooth basis $\mathbf{\Phi}$, $\mathcal{B} = \{\phi_k\}_{k=1}^K$ be a set of K distinct temporal basis functions, where $K \ll T$. and construct the Basis Projection Matrix $\mathbf{\Phi} \in \mathbb{R}^{T \times K}$

$$\mathbf{\Phi} = \begin{bmatrix} \phi_1(1) & \phi_2(1) & \dots & \phi_K(1) \\ \phi_1(2) & \phi_2(2) & \dots & \phi_K(2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(T) & \phi_2(T) & \dots & \phi_K(T) \end{bmatrix} \quad (5)$$

We define a learnable weight matrix $\mathbf{W} \in \mathbb{R}^{K \times D}$ such that for a $w_{k,m}$ it represents the contribution of the basis function k to sensor m . The perturbations $\mathbf{\Delta}(\mathbf{W})$ is defined by the linear projection.

$$\mathbf{\Delta}(\mathbf{W}) = \mathbf{\Phi} \mathbf{W} \quad (6)$$

So From Equation 4 and Equation 6 the counterfactual parameterized by $\mathbf{W}$ is

$$\mathbf{X}_{\text{cf}} = \Pi_{\mathcal{S}}(\mathbf{X} + \mathbf{\Phi} \mathbf{W}). \quad (7)$$

The operator $\Pi_{\mathcal{S}}$ strictly enforces mutability masks (preventing edits on immutable or contextual features), feature-wise minimum and maximum bounds,

and static-feature consistency over time. The feature schema ($\mathcal{S}$) serves as a rigorous specification of domain knowledge and physical constraints for the multivariate time series. It explicitly defines the role of each variable, distinguishing between actionable features that can be modified and those that are strictly immutable.

To generate a diverse set of N counterfactuals for a single query, we jointly optimise a collection of coefficient matrices $\mathcal{W} = \{\mathbf{W}^{(1)}, \dots, \mathbf{W}^{(N)}\}$. The overall loss function $\mathcal{J}(\mathcal{W})$ balances predictive validity with several competing regularisation terms to ensure plausibility:

$$\mathcal{J}(\mathcal{W}) = \lambda_v \mathcal{L}_{\text{valid}} + \rho \mathcal{L}_{\text{valid}}^2 + \rho (\mathcal{L}_{\text{cleanup}}), \quad (8)$$

where $\rho > 0$ is a dynamic augmented penalty coefficient that grows if the optimization stagnates, and c is a binary or fractional scaling factor used to down-weight the cleanup terms until the search discovers an initially valid region. The cleanup objective is defined as:

$$\mathcal{L}_{\text{cleanup}} = \lambda_p \mathcal{L}_{\text{prox}} + \lambda_c \mathcal{L}_{\text{coef}} + \lambda_g \mathcal{L}_{\text{group}} + \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_d \mathcal{L}_{\text{div}} + \lambda_{\text{sl}} \mathcal{L}_{\text{state}}. \quad (9)$$

The **Validity** Loss $\mathcal{L}_{\text{valid}}$ ensures that the counterfactuals generated achieve the desired prediction. For a regression task, the target output can be set to an interval of $y_{\text{target}} \in [y_{\min}, y_{\max}]$

$$\mathcal{L}_{\text{valid}} = \frac{1}{N} \sum_{i=1}^N (\max(0, y_{\min} - f(\mathbf{X}_{cf}))^2 + \max(0, f(\mathbf{X}_{cf}) - y_{\max})^2) \quad (10)$$

The **Proximity** Loss ($\mathcal{L}_{\text{prox}}$) is to penalise large deviations from the original input, encouraging minimal, actionable changes. To achieve this, we use our proximity loss, similar to that of Wachter [29], so for each feature d in D , the m_d^{-1} is generally set to 1

$$\text{med}_d = \text{median}_j(x_{j,d}), \quad \text{MAD}_d = \text{median}_j(|x_{j,d} - \text{med}_d|) \quad (11)$$

$$m_d^{-1} = \frac{1}{\text{MAD}_d + \epsilon} \quad (12)$$

From Equation 12 the Proximity Loss is

$$\mathcal{L}_{\text{prox}} = \frac{1}{N} \sum_{i=1}^N \left\| \Phi \mathbf{W}^{(i)} m_d^{-1} \right\|_F^2 \quad (13)$$

The **Coefficient Sparsity** Loss ($\mathcal{L}_{\text{coef}}$) encourages compact counterfactual explanations in the basis space, we penalise the ℓ_1 -norm of the basis coefficients.

$$\mathcal{L}_{\text{coef}} = \frac{1}{N} \sum_{i=1}^N \left\| \mathbf{W}^{(i)} \right\|_1 \quad (14)$$

Rather than applying pointwise sparsity in the time domain, we apply a Group Lasso penalty across predefined sets of actionable features $\mathcal{G} = \{g_1, \dots, g_M\}$. The **Channel Group Sparsity** ($\mathcal{L}_{\text{group}}$) ensures the explanation acts on the smallest necessary number of distinct sensor channels:

$$\mathcal{L}_{\text{group}} = \frac{1}{N} \sum_{i=1}^N \sum_{g \in \mathcal{G}} \sqrt{\sum_{t=1}^T \sum_{d \in g} (\Delta_{t,d}^{(i)})^2}. \quad (15)$$

The **Smoothness Loss** ($\mathcal{L}_{\text{smooth}}$) regularises temporal irregularities; it penalises rapid and sudden temporal changes in the counterfactual perturbation in Equation 6 by measuring its discrete curvature (second difference) over time. Concretely, for each counterfactual i and feature d , $\Delta^{(i)} = \Phi \mathbf{W}^{(i)}$, then the second order derivative of Δ also known as the Roughness Penalty [11].

$$\nabla^2(\Delta^{(i)})_{t,d} = \Delta_{t+1,d}^{(i)} - 2\Delta_{t,d}^{(i)} + \Delta_{t-1,d}^{(i)} \quad (16)$$

$$\mathcal{L}_{\text{smooth}}(\mathcal{W}) = \frac{1}{N(T-2)D} \sum_{i=1}^N \sum_{t=2}^{T-1} \sum_{d=1}^D (\Delta_{t+1,d}^{(i)} - 2\Delta_{t,d}^{(i)} + \Delta_{t-1,d}^{(i)})^2 \quad (17)$$

The **Diversity Loss** via Determinantal Point Processes(DPP) ($\mathcal{L}_{\text{dpp}}$) is to shape diverse trajectories of N counterfactuals. DPP ensures that the set of counterfactuals is non-redundant [18] [23].

$$\hat{\mathbf{v}}_i = \frac{\text{vec}(\mathbf{W}^{(i)})}{\|\text{vec}(\mathbf{W}^{(i)})\|_2} \quad (18)$$

$$S_{i,j} = \hat{\mathbf{v}}_i^T \hat{\mathbf{v}}_j \quad (19)$$

$$\mathcal{L}_{\text{dpp}}(\mathcal{W}) = -\log(\det(\mathbf{S} + \epsilon \mathbf{I})) \quad (20)$$

In real-world time series, actionable recourse should ideally target manipulable *action* features rather than endogenous *state* variables. To prevent optimisation infeasibility while still discouraging state alterations, our feature schema $\mathcal{S}$ applies a soft ℓ_1 -norm penalty, the State Lock Loss ($\mathcal{L}_{\text{state}}$), to perturbations strictly within the state-feature subset $\mathcal{D}_{\text{state}} \subseteq \{1, \dots, D\}$

$$\mathcal{L}_{\text{state}} = \frac{1}{NT|\mathcal{D}_{\text{state}}|} \sum_{i=1}^N \sum_{t=1}^T \sum_{d \in \mathcal{D}_{\text{state}}} |\Delta_{t,d}^{(i)}|. \quad (21)$$

By incorporating this term into the overall cleanup objective with a weight λ_{sl} , the optimization framework is encouraged to prioritize recourse through actionable channels, resorting to state modifications only when strictly necessary to cross the target decision boundary.

Thus, the overall optimisation problem seamlessly balances validity, continuous schema constraints, and temporal plausibility in the basis space:

$$\mathcal{W}^* = \arg \min_{\mathcal{W}} \mathcal{J}(\mathcal{W}), \quad \mathbf{X}_{\text{cf}}^{(i)*} = \Pi_S(\mathbf{X} + \Phi \mathbf{W}^{(i)*}), \quad i = 1, \dots, N. \quad (22)$$

Our Algorithm 1 generates counterfactual time series by optimising a low-dimensional coefficient tensor $\mathbf{W}$ instead of perturbing each time step independently. For any chosen Basis family we first construct a temporal basis matrix Φ , then we project the $\mathbf{W}$ into the input space $\Delta = \Phi \mathbf{W}$, next we form the candidate counterfactuals as $\mathbf{X}_{\text{cf}} = \Pi_S(\mathbf{X} + \Delta)$, where Π_S enforces mutability masks, bounds, and optional structural constraints. The optimisation objective combines a validity term with proximity, coefficient sparsity, group/channel sparsity, smoothness, and diversity penalties; we implement the validity as an augmented-penalty style through $\lambda_v \mathcal{L}_{\text{valid}} + \rho \mathcal{L}_{\text{valid}}^2$. The optimisation is performed over multiple restarts, using Adam [16] for the main search and optional LBFGS[19] refinement, and the final counterfactual set is selected by a validity-first lexicographic criterion: valid solutions are preferred first, and ties are broken by validity error, proximity, group sparsity, and smoothness.

Algorithm 1 Basis-Guided Counterfactual Generation

Require: Predictor f , query $\mathbf{X} \in \mathbb{R}^{T \times D}$, target $\mathcal{T}$, schema $\mathcal{S}$, basis Φ , CF count N .

Ensure: Optimal counterfactual set $\mathbf{X}_{\text{cf}}^*$.

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1: Init global best  $K_{\text{global}} \leftarrow (\infty, \dots, \infty)$ ,  $\mathbf{X}_{\text{cf}}^* \leftarrow \text{null}$ , construct  $\Phi$ .
2: for restart  $r = 1 \dots R$  do
3:   Init  $\mathbf{W}$  (zeros if  $r = 1$ , else  $\mathcal{N}(0, \sigma_{\text{init}})$  shifted by previous best).
4:    $\rho \leftarrow \rho_{\text{init}}$ ,  $p \leftarrow 0$ ,  $K_{\text{local}} \leftarrow (\infty, \dots, \infty)$ ,  $\text{ever\_valid} \leftarrow \text{False}$ .
5:   for step  $s = 1 \dots S_{\text{Adam}}$  do ▷ Phase 1: Adam Search
6:      $c \leftarrow 1.0$  if  $\text{ever\_valid}$  else  $c_{\text{prevalid}}$ 
7:      $\mathbf{X}_{\text{cf}}, \Delta \leftarrow \Pi_S(\mathbf{X} + \Phi \mathbf{W})$  ▷ Continuous schema constraints
8:      $\mathcal{J} \leftarrow \lambda_v \mathcal{L}_{\text{valid}}(f(\mathbf{X}_{\text{cf}})) + \rho \mathcal{L}_{\text{valid}}^2 + c \mathcal{L}_{\text{cleanup}}$ 
9:     Update  $\mathbf{W}$  via Adam step minimizing  $\mathcal{J}$ .
10:     $K \leftarrow (\text{invalid}, \mathcal{L}_{\text{valid}}, \mathcal{L}_{\text{prox}}, \mathcal{L}_{\text{group}}, \mathcal{L}_{\text{smooth}})$ 
11:    if  $K < K_{\text{local}}$  then  $K_{\text{local}} \leftarrow K$ ,  $\mathbf{W}_{\text{local}}^* \leftarrow \mathbf{W}$ ,  $p \leftarrow 0$  else  $p \leftarrow p + 1$ 
12:    if  $\max(\text{violation}) \leq \epsilon$  then  $\text{ever\_valid} \leftarrow \text{True}$ 
13:    if  $\neg \text{ever\_valid}$  and  $s \bmod 25 == 0$  then  $\rho \leftarrow \min(\rho \cdot \rho_{\text{growth}}, \rho_{\text{max}})$ 
14:    if  $\text{ever\_valid}$  and  $p \geq p_{\text{max\_patience}}$  then break
15:  end for
16:  if  $S_{\text{LBFGS}} > 0$  then ▷ Phase 2: L-BFGS Refinement
17:    Refine  $\mathbf{W}_{\text{local}}^*$  via L-BFGS; update  $K_{\text{local}}, \mathbf{W}_{\text{local}}^*$  if improved.
18:  end if
19:   $\mathbf{X}_{\text{cf}}^{\text{final}} \leftarrow \Pi_S^{\text{final}}(\mathbf{X} + \Phi \mathbf{W}_{\text{local}}^*)$  ▷ Final discrete constraints
20:  if  $K_{\text{local}} < K_{\text{global}}$  then  $K_{\text{global}} \leftarrow K_{\text{local}}$ ,  $\mathbf{X}_{\text{cf}}^* \leftarrow \mathbf{X}_{\text{cf}}^{\text{final}}$ 
21: end for
22: return  $\mathbf{X}_{\text{cf}}^*$ 

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Table 1. Summary of the datasets used in the experiments.

Dataset	Task	#Features	Sequence Length
NASA CMAPSS FD001	RUL regression	13	50
NASA CMAPSS FD002	RUL regression	6	50
NASA CMAPSS FD003	RUL regression	11	50
NASA CMAPSS FD004	RUL regression	4	50
IEEE PHM 2012 PRONOSTIA	RUL regression	8	256
AEP	Forecasting	21	12

4 Experiments and Results

We evaluate the proposed basis-guided framework on three multivariate time-series benchmarks: NASA CMAPSS and IEEE PHM 2012 PRONOSTIA for Remaining Useful Life (RUL) prognostics, and the AEP dataset for electricity demand forecasting. This diverse selection allows us to rigorously assess the framework’s ability to generate valid, sparse, smooth, and diverse counterfactuals across varying temporal dynamics.

4.1 Datasets

First dataset is *NASA CMAPSS* [27], it is a standard turbofan engine degradation dataset; we formulate the task as Remaining Useful Life (RUL) regression across all four sub-datasets (FD001–FD004) using a 50-step lookback window. Next, *IEEE PHM 2012 PRONOSTIA* [2] provides accelerated run-to-failure bearing degradation data, which we similarly frame as an RUL regression task based on vibration measurements. Finally, for forecasting, we utilise the *Appliances Energy Prediction* (AEP) dataset [8] to predict future residential appliance energy consumption from 10-minute-resolution multivariate sensor readings.

4.2 Predictive Models

We use an encoder-only Transformer as our fixed blackbox model for all the datasets. Each input sequence $\mathbf{X} \in \mathbb{R}^{T \times D}$ is first projected from the original feature space to a latent representation of dimension d_{model} through a linear input layer, after which sinusoidal positional encodings are added to preserve temporal order. The resulting sequence is processed by a stack of L Transformer encoder layers with multi-head self-attention and position-wise feed-forward sub-layers, and the hidden representation of the final time step is passed through a small fully connected regression head to produce the scalar prediction. This design follows the standard Transformer encoding principle of modelling long-range temporal dependencies through self-attention while remaining fully parallelisable across time steps.

Table 2. Performance of the predictive models before counterfactual generation.

Dataset	Model	MAE ↓	RMSE ↓
NASA CMAPSS FD001	Transformer	9.3737	13.1924
NASA CMAPSS FD002	Transformer	11.5403	16.6911
NASA CMAPSS FD003	Transformer	9.9746	14.9073
NASA CMAPSS FD004	Transformer	11.9564	19.7240
IEEE PHM 2012 PRONOSTIA	Transformer	0.2574	0.3314
AEP	Transformer	0.0716	0.0716

4.3 Counterfactual Generation and Evaluation

We compare the proposed basis-guided generator with two optimisation-based baselines, CoMTE and ForecastCF. We focus on optimisation-based baselines because they operate on the input sequence of the fixed predictor and therefore avoid the additional reconstruction error introduced by latent generative explainers. For the proposed method, perturbations are expressed as $\Delta = \Phi W$, where $\Phi \in \mathbb{R}^{T \times K}$ is a temporal basis matrix and W contains the learned feature-wise coefficients. We evaluate five basis families: B-spline [6], Gaussian RBF [7], Ricker wavelet [26], Legendre polynomial [1], and Fourier [17]. Unless otherwise stated, each query produces three counterfactuals and the predictive model remains frozen.

For each test query, we define a desired target prediction and generate a valid and diverse set of counterfactual trajectories by optimising the basis coefficients while keeping the predictive model fixed. To strictly enforce physical domain constraints, the feature schema $\mathcal{S}$ normalises all variables to the $[0, 1]$ interval and clamps perturbations during the search. Furthermore, state-variable edits are strictly prohibited; the optimiser is restricted to modifying only the designated *action* features to ensure the resulting recourse is operationally actionable.

For a query sequence $\mathbf{X}$, the target is specified as a regression band rather than a single point. For RUL datasets, the counterfactual target is a healthier operating state with a higher predicted RUL; for AEP, the target is a lower predicted energy consumption. The feature schema normalises variables to $[0, 1]$ and clamps generated sequences to this range. The editable roles are dataset dependent: CMAPSS and IEEE restrict edits to action features, whereas AEP allows both action and environmental state features to vary while keeping context variables locked. This distinction is important for interpreting the AEP heatmap in Fig. 3.

The hyperparameters were not identical across datasets. Table 3 summarises the settings used for the common-format evaluation in Table 4; the later AEP energy-reduction diagnostic used a stricter 200-query setting with $K = 10$, $\lambda_v = 12.0$, $\lambda_p = 8.0$, $\lambda_c = 0.5$, $\lambda_s = 1.5$, $\lambda_d = 0$, learning rate 10^{-3} , and 1000 iterations.

Table 3. Counterfactual generation settings used in the common-format evaluation. The loss tuple is $(\lambda_v, \lambda_p, \lambda_c, \lambda_g, \lambda_s, \lambda_d)$.

Dataset	Queries	Basis sizes K	Editable roles	Optimisation settings
CMA PSS	20	{4, 6, 8, 10}	action	lr 0.02, 500 iters, 8 restarts, (1.0, 0.50, 0.02, 0.15, 0.25, 0.20)
IEEE PRONOSTIA	5	{4, 6, 8, 10}	action	lr 0.02, 300 iters, 5 restarts, (1.0, 0.50, 0.02, 0.15, 0.25, 0.20)
AEP	20	{4, 6, 8, 10}	action+state	lr 0.02, 500 iters, 6 restarts, (1.0, 0.35, 0.02, 0.15, 0.25, 0.20)

Quantitative Evaluation Table 4 reports the two baselines and the best-performing basis-guided variant for each dataset, selected primarily by validity and then by the remaining quality terms. The omitted basis rows were still evaluated during model selection.

Table 4. Common-format counterfactual-quality comparison. For each dataset, the table shows both baselines and the selected best basis-guided row.

Dataset	Method	Validity ↓ / Success ↑	Proximity ↓	Sparsity ↓	Smoothness ↓	Diversity ↑
NASA CMAPSS FD001	CoMTE	0.016410 / 0.650	0.000178	1.115979	0.000075	0.039705
	ForecastCF	0.022332 / 0.600	0.000167	1.120008	0.000104	0.058820
	Ours(Polynomial)	0.000000 / 1.000	0.000134	0.727856	0.000002	0.127879
NASA CMAPSS FD002	CoMTE	0.010654 / 0.400	0.000122	0.443792	0.000077	0.045463
	ForecastCF	0.021707 / 0.300	0.000132	0.502939	0.000089	0.048989
	Ours(Wavelet)	0.000031 / 0.800	0.000623	0.800890	0.000198	0.012504
NASA CMAPSS FD003	CoMTE	0.012224 / 0.850	0.000175	0.960334	0.000058	0.096041
	ForecastCF	0.013502 / 0.700	0.000165	0.949651	0.000079	0.097964
	Ours(RBF)	0.000000 / 1.000	0.000097	0.452238	0.000000	0.050578
NASA CMAPSS FD004	CoMTE	0.000815 / 0.600	0.000085	0.288181	0.000068	0.128294
	ForecastCF	0.001981 / 0.600	0.000088	0.291157	0.000105	0.217544
	Ours(Polynomial)	0.000000 / 1.000	0.000031	0.114515	0.000001	0.000557
IEEE PRONOSTIA	CoMTE	0.000754 / 0.800	0.000459	1.302113	0.000709	0.263627
	ForecastCF	0.000237 / 0.800	0.000454	1.325139	0.000709	0.268268
	Ours(Wavelet)	0.000000 / 1.000	0.000422	0.818159	0.000633	0.626258
AEP	CoMTE	0.000036 / 0.950	0.000219	0.898315	0.000445	0.060877
	ForecastCF	0.000011 / 0.950	0.000216	0.868407	0.000450	0.060376
	Ours(Polynomial)	0.000000 / 1.000	0.000005	0.056649	0.000003	0.000945

Validity. The basis-guided methods are most consistent on the prognostics datasets. On CMAPSS FD001, FD003, and FD004, the selected basis rows reach zero target-band violation and 1.000 success, while CoMTE and ForecastCF remain between 0.600 and 0.850 success. IEEE PRONOSTIA follows the same pattern, with the Wavelet basis reaching 1.000 success compared with 0.800 for both baselines. FD002 is the hardest CMAPSS subset: the Wavelet basis gives the smallest residual validity loss, but its success rate is 0.800, while other basis families in the search reached 0.950 success at higher residual cost. Thus, the result is not a blanket claim that every basis wins every metric; rather, the basis parameterisation makes target-reaching substantially more reliable on most RUL cases.

Proximity and Sparsity. Polynomial and RBF bases usually provide the most compact interventions. On FD003, RBF reduces group sparsity from 0.960334

and 0.949651 for CoMTE and ForecastCF to 0.452238, while also improving proximity to 0.000097. On FD004, Polynomial reduces group sparsity to 0.114515, roughly one third of the baseline values. The AEP common-format run shows the strongest sparsity gain: Polynomial reaches zero validity loss with group sparsity 0.056649, compared with 0.898315 for CoMTE and 0.868407 for ForecastCF. In the separate AEP energy-reduction diagnostic with a stricter 20 Wh success threshold, the baselines obtain higher hit rates, but they do so under a different objective and sparsity definition; we therefore treat AEP as a trade-off case rather than as a uniform win.

Smoothness and Diversity. The low-dimensional basis strongly suppresses high-frequency perturbations. RBF, Polynomial, and Fourier reach near-zero smoothness penalties on several CMAPSS subsets, and the IEEE Wavelet row improves smoothness from 0.000709 for both baselines to 0.000633. Diversity is more nuanced. Baselines sometimes report larger pairwise diversity, but this can arise from point-wise fluctuations or from invalid solutions. The DPP term in our method instead searches for diversity within the smooth coefficient space, so diversity should be interpreted jointly with validity and smoothness.

4.4 Qualitative and Trade-off Analysis

Trajectory Plausibility and Diversity. Figure 1 visualises a CMAPSS FD003 test instance. In the full-cycle view, the B-spline counterfactual modifies only the selected 50-cycle window and follows the surrounding degradation trend instead of inserting isolated point-wise jumps. In the window-level view, the three generated counterfactuals move the model prediction from an original RUL of 82 toward the target band around 100; the displayed counterfactual predictions are approximately 98. The trajectories are not visually identical on the most affected channels, yet they remain smooth and close to the original signal, which is the intended trade-off between diversity and physical plausibility.

IEEE Bearing Perturbations. Figure 2 shows the corresponding perturbation footprint for IEEE PRONOSTIA. The B-spline counterfactual remains smooth over the 256-step bearing window and concentrates changes on vibration amplitude and spectral statistics, while mean channels remain nearly unchanged. This provides a qualitative view of the same behaviour observed in Table 4: the basis space can reach the desired RUL region without introducing scattered point-wise corrections.

Perturbation Localisation. Figure 3 shows the AEP perturbation footprint. The largest magnitude changes occur on the **Appliances** channel, especially in the later time steps, but the heatmap also contains smaller compensating changes across temperature, humidity, and weather variables. This is expected because the AEP configuration used for this figure allowed state edits. The figure, therefore, illustrates a useful forecasting trade-off: allowing state variables can produce coherent low-energy scenarios, but it is less directly actionable than the action-only setting used for RUL recourse.

Validity vs. Proximity. Figure 4 summarises the CMAPSS FD003 trade-off. The basis methods lie on the zero-validity axis, showing that they reach the

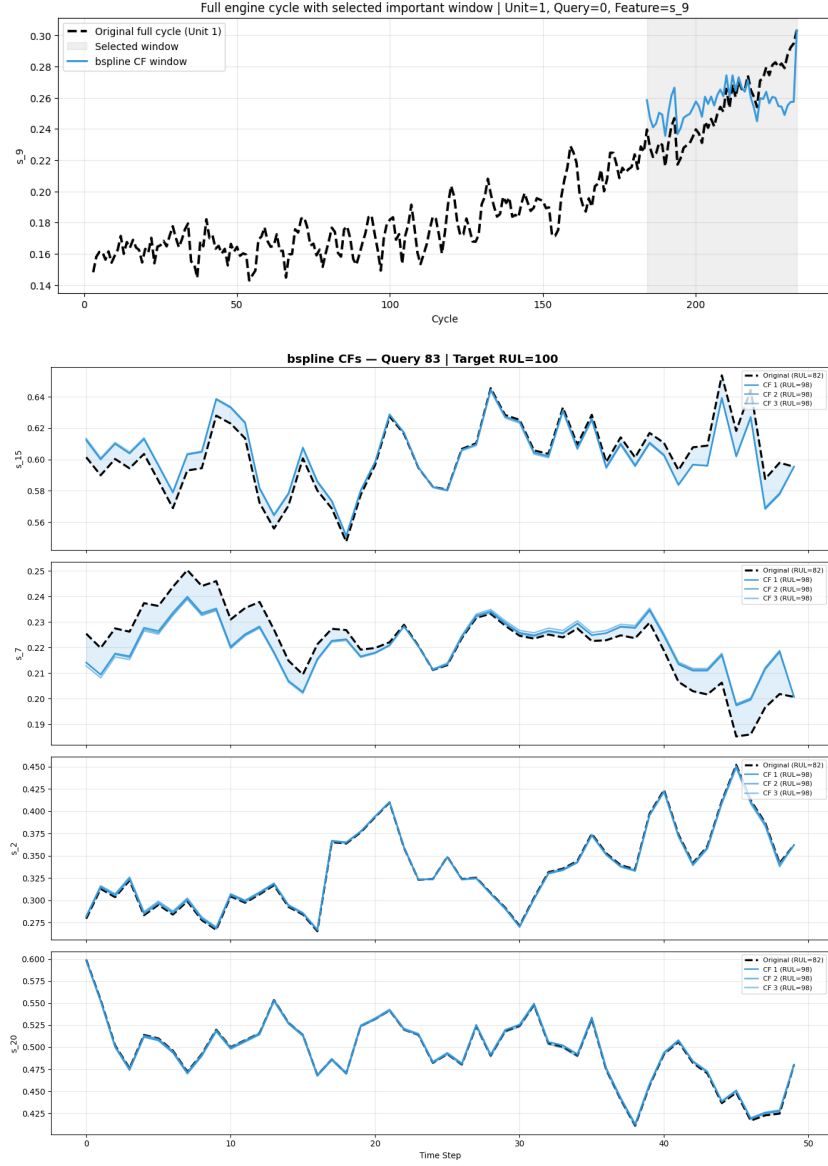


Fig.1. Macroscopic and microscopic evaluation of counterfactual plausibility on CMAPSS FD003. (Top) The B-spline counterfactual is confined to the selected look-back window while following the larger engine-cycle trend. (Bottom) Three smooth counterfactual trajectories for query 83 move the predicted RUL from 82 into the target region near 100.

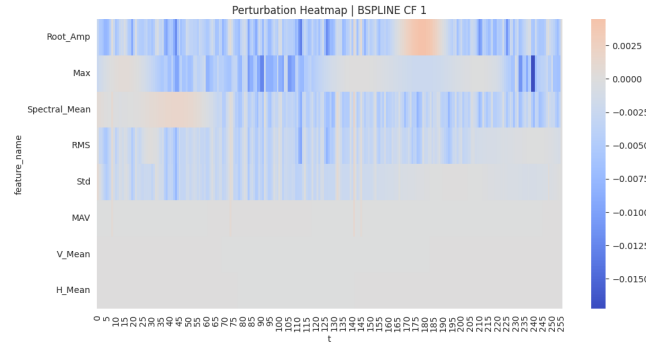


Fig. 2. IEEE PRONOSTIA perturbation heatmap for a B-spline counterfactual. The edits are smooth over the 256-step bearing window and concentrated on vibration-amplitude and spectral statistics, while mean channels remain nearly unchanged.

target band. Among them, RBF and Polynomial are closest to the origin, while Wavelet remains valid but uses a larger perturbation. CoMTE and ForecastCF have small proximity values, but their non-zero validity losses show that the smaller edits often fail to reach the desired RUL region. This supports the main conclusion that the basis space helps the optimiser find target-reaching changes without resorting to temporally jagged corrections.

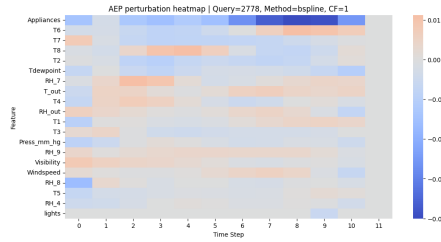


Fig. 3. AEP perturbation heatmap for a B-spline counterfactual. The dominant change is on **Appliances**, with smaller state-feature adjustments due to the editable-state AEP setup.

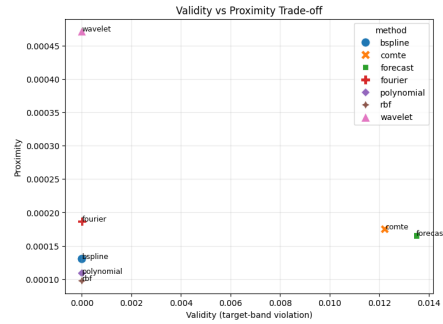


Fig. 4. Validity-proximity trade-off on CMAPSS FD003. Basis methods reach the target band; RBF and Polynomial provide the most proximal valid solutions.

5 Conclusion

In this paper, we introduced a basis-guided framework for generating counterfactual explanations for multivariate time series. By parameterising perturbations in a low-dimensional temporal basis space, the method embeds temporal coherence directly into the optimisation while still balancing validity, proximity, sparsity, smoothness, and diversity. Across NASA CMAPSS, IEEE PRONOSTIA, and AEP, basis-guided counterfactuals usually achieve stronger target validity and group sparsity than point-wise baselines, with the most consistent gains on the RUL tasks.

The main trade-off is computational: basis-space optimisation requires multiple gradient steps and random restarts, but this cost is acceptable in predictive-maintenance and forecasting settings where explanation quality is more important than millisecond inference. A remaining limitation is causal realism: action-only schemas prevent implausible direct edits to state variables, but physical interventions can still produce downstream state changes. Future work will integrate Structural Causal Models (SCMs) into the schema so counterfactuals can simulate such effects rather than only constrain them.

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